

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: I am looking for a type c connector material

Assistant: To better assist you, could you please clarify:

1. What specific part of the Type-C connector will the polymer be used for? (For example: housing, insulation, overmold/strain relief, or cosmetic bezel)
2. What is the end-use application of this connector? (For instance: Is it for a laptop, smartphone, game console, or another device?)

This will help me understand your requirements more precisely and suggest suitable material options.

User: it is in consumer electronics - home telecommunication hubs

Assistant: Thank you for clarifying the end-use application. To proceed, could you please provide the following details:

1. What is the typical operating environment for the connector in your home telecommunication hub? (For example: indoor, exposure to dust or humidity, visible to users, etc.)
2. What is the minimum wall thickness of the polymer part you are considering?
3. Do you have any preferred base materials for this application (such as specific polymers you have used before)?
4. Are you open to alternative material suggestions if there is no strict preference for a

particular polymer?

These details will help ensure the material recommendations fit your requirements.

User: It is will be used indoors - and I don't have a preferred material. thickness 0.4mm

Assistant: Here are the key material requirements (CTQs) assumed for your application. Please confirm if these are correct for your USB Type-C connector housing in a home telecommunication hub:

1. Minimum wall thickness: 0.4 mm
2. Flammability classification (UL 94): V-0 at 0.4 mm
3. Comparative Tracking Index (CTI): 600 V
4. Lead-free soldering resistance (reflow up to 260–280°C)
5. Non-halogenated/red phosphorous free flame retardant
6. High flow for thin-wall molding
7. Improved weld line strength
8. Dimensional stability
9. Electrical insulation
10. Injection molding processing
11. High gloss aesthetics
12. Relative Thermal Index – Electrical (RTI Elec): 130–140°C at 0.4 mm

Are all of these requirements correct for your application?

User: looks good recommend the grades

Assistant: Here are potential material options for your USB Type-C connector housing in a home telecommunication hub. All grades below meet your requirements for thin-wall molding (0.4 mm), UL 94 V-0 flammability, CTI 600 V, lead-free soldering resistance,

dimensional stability, and good aesthetics:

1. Zytel® HTNFR42G30NH NC010 (Rank 1): Balanced stiffness, high flow, natural color, RTI Elec 130°C at 0.4 mm.
2. Zytel® HTNFR52G30NH NC010 (Rank 2): Higher RTI Elec (140°C), natural color, high flow, 30% GF.
3. FRIANYL® XT4 GF30 V0I NC 1102/F (Rank 3): PPA, 30% GF, natural color, RTI Elec 140°C, heat stabilized.
4. Zytel® HTNFR52G45NHF NC010 (Rank 4): Maximum stiffness with 45% GF, natural color, RTI Elec 140°C.
5. Zytel® HTNFR52G35NHF BK337 (Rank 5): Black variant, 35% GF, strong historical use in Type-C connectors, RTI Elec 140°C.

These grades may be suitable for your application, but final material selection and product design decisions should be based on your specific requirements and testing. Please ensure compliance with all applicable standards and regulations. Celanese does not manufacture or certify products for these applications.

*Note: This recommendation is generated by an AI assistant and should be considered purely informational.*